Practical no. 7A

Aim: Develop a PWA that works in offline mode using Service Worker.

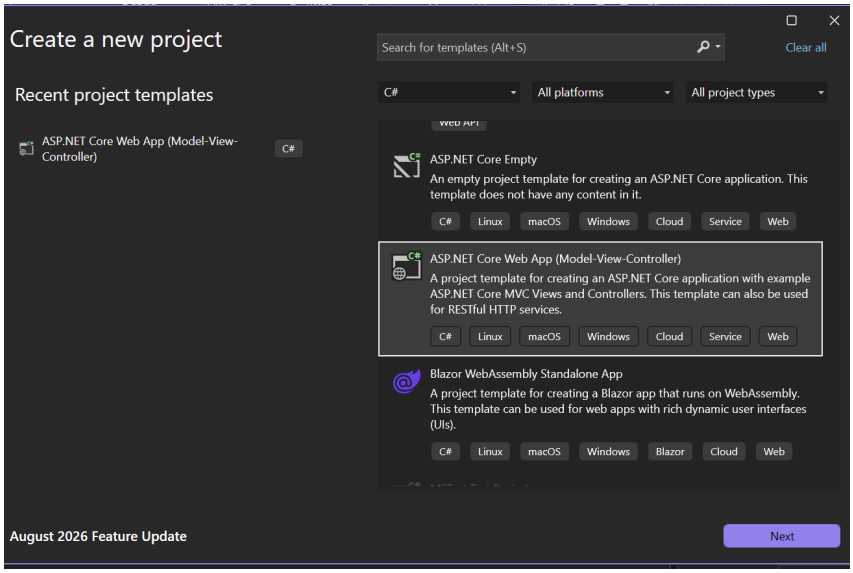
Software Required:

Visual Studio 2026

Procedure:

STEP 1: CREATE NEW PROJECT

1. Open Visual Studio 2026.
2. Click: Create a New Project
3. Open Visual Studio 2026.
4. Select Create a new project.
5. Search for ASP.NET Core Web App (Model-View-Controller).
6. Click Next.
7. Set the Project Name to Practical No. 6b.
8. Select a suitable location and click Next.
9. Select the installed/latest .NET version and keep the default options.
10. Click Create.



STEP 2: CREATE HOME PAGE

Open Views → Home → Index.cshtml and replace its contents with:

```
@{
    ViewData["Title"] = "Offline PWA";
}
```

```
<div class="text-center">

    <h1>Offline PWA</h1>

    <h2>Student Information Portal</h2>

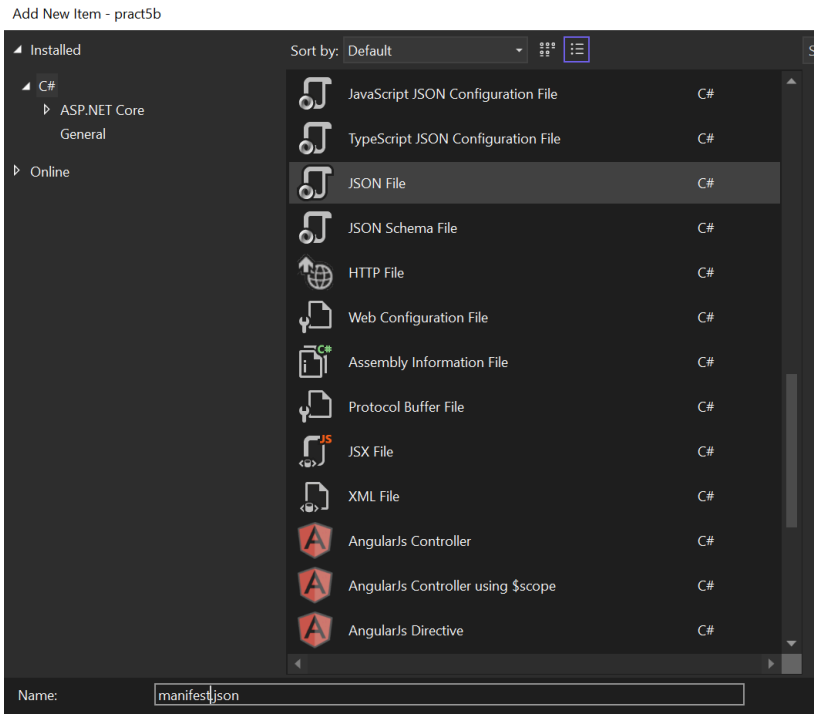
    <p>Welcome to the Student Information Portal.</p>

    <p>This page can also be accessed offline.</p>

</div>
<script src="~/js/site.js"></script>
```

Step 3: Create Manifest File

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select json file.
4. Enter File Name: **manifest.json**
5. Click Add.



Write the following code in the manifest.json file.

manifest.json

```
{

  "name": "Offline PWA",
   "short_name": "OfflinePWA",
   "start_url": "/",
   "display": "standalone",
   "background_color": "#ffffff",
   "theme_color": "#007bff"

}
```

Step 4: Create Service Worker

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select Javascript file.
4. Enter File Name: **service-worker.js**
5. Click Add.

```
const CACHE_NAME = "my-pwa-cache";

self.addEventListener("install", function (event) {

   event.waitUntil(
      caches.open(CACHE_NAME).then(function (cache) {

         return cache.add("/");

      })
   );

});

self.addEventListener("fetch", function (event) {

   event.respondWith(

      caches.match(event.request).then(function (response) {
```

```
        if (response) {
            return response;
        }

        return fetch(event.request);

    })

  );

});
```

Step 5: Register the Service Worker
Open:

wwwroot → js → site.js

Add:

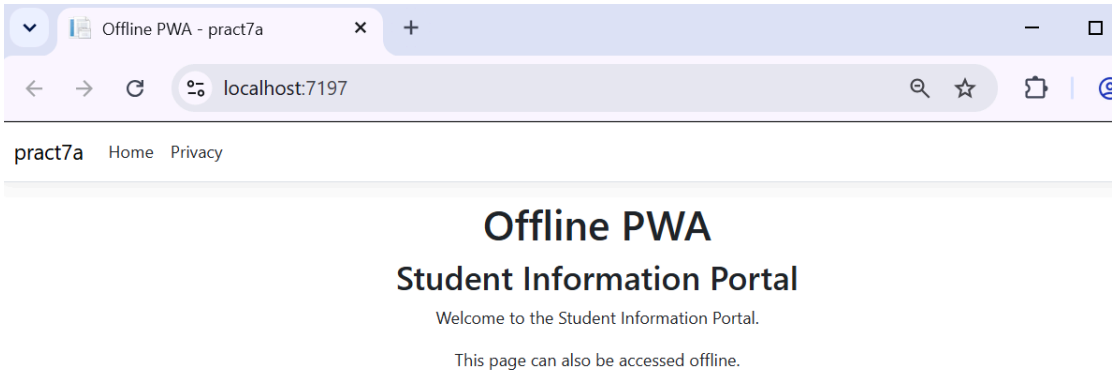
```
if ("serviceWorker" in navigator) {

  navigator.serviceWorker.register("/service-worker.js");

}
```

Step 6: Run the Application

Press F5



Step 7: Check the Service Worker
Press F12 → Application → Service Workers.

Check that service-worker.js is registered and running.

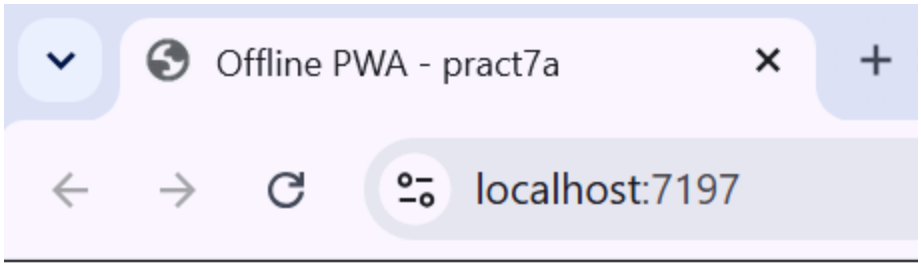
Step 8: Check Cache Storage

In Developer Tools, go to: Application→Cache Storage→Click: my-pwa-cache.

You should see: /

Step 10: Test Offline Mode

1. Press: F12
2. Open: Network Tab
3. Select: Offline
4. Refresh page.



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- Home
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Offline PWA

Student Information Portal

Welcome to the Student Information Portal.

This page can also be accessed offline.

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